

Mixed-Strategy Equilibrium

John Van Huyck

1

Introduction

- Are people good at being unpredictable?
 - In many situations, the unique Nash equilibrium requires players to select a pure strategy in an unpredictable way
 - That is, the only mutually consistent strategy combination is a mixed strategy equilibrium
- Examples
 - Penalty kick in soccer
 - Serve in tennis
 - Strategic bombing
 - Inspecting potential polluters

2



Constant-sum MSE games

- Do not have some of complications of other games
 - It is not possible for one player to help another without hurting herself, so social preferences and reciprocation play no role.
 - The games in Chapter 3 have a unique Nash equilibrium.
- Moreover, the constant-sum games have a unique Nash equilibrium in the players' maximin strategies.
 - Such games have a value and playing the maximin strategy guarantees the player will earn at least the value of the game.
 - Von Neumann and Morgenstern's prescriptive theory suggests playing a maximin strategy in constant-sum games.

3



Maximin strategies

- The maximin strategy maximizes the minimum a player can earn.
- This produces a Nash equilibrium in zero-sum games because players' interests are strictly opposed.
- If others do whatever they can to get the most, their actions will give me the least, so I should try to maximize the least I can get.
- Since maximin has a compelling prescriptive theory and coincides with Nash equilibrium, there is a good chance that the Nash equilibrium will predict what players do in these games.

4

O'Neill (1987)

- O'Neill revived interest research on zero-sum games
- Restricted payoff to two outcomes, a 5 cent win or loss, to avoid problems with risk aversion.
- Subjects played in fixed pairs
- 105 repeated pairings

5

O'Neill (1987)

Table 3.3. Row player payoffs (in cents) and results in a 4×4 game

Row choice	Column player				MSE probability	Actual frequency	QRE estimate
	1	2	3	J			
1	-5	5	5	-5	0.20	0.221	0.213
2	5	-5	5	-5	0.20	0.215	0.213
3	5	5	-5	-5	0.20	0.203	0.213
J	-5	-5	-5	5	0.40	0.362	0.360
MSE probability	0.20	0.20	0.20	0.40			
Actual frequency	0.226	0.179	0.169	0.426			
QRE estimate	0.191	0.191	0.191	0.426			

Sources: O'Neill (1987).

Note: QRE estimate is $\hat{\lambda} = 1.313$.

6

Brown and Rosenthal (1990)

- Aggregate analysis misses the fact that players are not mixing independently using the MSE probabilities.
- Row players are predicted to win 40.0 percent of the time in the MSE
 - On average actually win 41.0 percent of the time
 - But Brown and Rosenthal (1990) note that win rates across specific pairs range from 30 percent to 54 percent.
 - Much of the variation in win rates comes from correlation between players' choices.

7

Temporal Dependence

Table 3.4. Tests for temporal dependence in zero-sum games

Effect	Coefficient	Percentage of players with significant ($p < .05$) effects			
		BR	RB exp. 1	RB exp. 2	BR94
Guessing	b_0	8	8	5	10
Previous opponent choices	b_1, b_2	30	42	35	12
Previous outcomes	c_1, c_2	38	28	20	6
Previous choices and outcomes	b_1, b_2, c_1, c_2	44	55	38	12
Previous own choices	a_1, a_2	48	48	42	42
All effects	a_i, b_i, c_i	62	72	50	52

Sources: Brown and Rosenthal (1990); Rapoport and Boebel (1992); Budescu and Rapoport (1994).

Notes: BR denotes Brown and Rosenthal (1990); RB denotes Rapoport and Boebel (1992); BR94 denotes Budescu and Rapoport (1994). $J_{t+1} = a_0 + a_1 J_t + a_2 J_{t-1} + b_0 J_{t+1}^* + b_1 J_t^* + b_2 J_{t-1}^* + c_1 J_t J_t^* + c_2 J_{t-1} J_{t-1}^*$.

8

Brown and Rosenthal (1990)

- For 62 percent of subjects can reject the null hypothesis of no temporal dependence.
- However, only 8 percent have a significant guessing parameter, so most subject were not able to use this information to guess what their opponent would do in the current period.
- Camerer suggests that the MSE can be seen as an equilibrium in beliefs rather than in mixtures.
- Although players are not mixing according to the MSE, opponents' departures are not being exploited.

9

Rapoport and Boebel (1992)

Table 3.5. Payoffs and results in 5×5 games

Row choice	Column choice					MSE probability	Actual frequency		QRE estimates	
	C	L	F	I	O		Experiment 1	Experiment 2	Experiment 1	Experiment 2
C	W	L	L	L	L	0.375	0.293	0.306	0.286	0.309
L	L	L	W	W	W	0.250	0.305	0.324	0.302	0.296
F	L	W	L	L	W	0.125	0.123	0.100	0.138	0.132
I	L	W	L	W	L	0.125	0.119	0.115	0.138	0.132
O	L	W	W	L	L	0.125	0.160	0.155	0.138	0.132
Equilibrium probability	0.375	0.250	0.125	0.125	0.125					
Actual frequency (experiment 1)	0.352	0.180	0.218	0.099	0.151					
Actual frequency (experiment 2)	0.346	0.193	0.202	0.116	0.143					
QRE estimate (experiment 1)	0.412	0.169	0.140	0.140	0.140					
QRE estimate (experiment 2)	0.410	0.184	0.135	0.135	0.135					

Sources: Rapoport and Boebel (1992).

Note: (W,L) payoffs are (\$10,-\$6) in experiment 1, (\$15,-\$1) in experiment 2. QRE estimate parameters are $\hat{\lambda}_1 = 0.248$, $\hat{\lambda}_2 = 0.327$.

10



Rapoport and Boebel (1992)

- Overall results replicate O'Neill's
- Actual frequencies are generally between MSE predictions and equal probability
- MSE predictions can be rejected by a Chi-square test for 85 percent of the subjects.
- Note that QRE estimates do an adequate job of capturing deviations from MSE, especially for row players.
- No persistent skill differences in these games, because the correlation between a single subject's win rates in his or her two different sessions is only 0.17.

11



Mookerjee and Sopher (1997)

- 40 periods
- A win is worth either 5 or 10 rupees
- Monthly room and board in student dorm is 600 rupees, so stakes are large.
- Again multinomial MSE prediction can be rejected, but the results are close to MSE.
- Weakly dominated strategies are rarely played (2-8 percent)
- Strategic equivalence from combining strategies 3-5 in game 2 into a single strategy in game 1 holds on average.

12

Mookerjee and Sopher (1997)

Table 3.6. Row payoffs and results in games 1 and 3

Row choice	Column choice				MSE probability	Actual frequency
	1	2	3	4		
1	W	L	L	W	0.375	0.318
2	L	L	W	W	0.250	0.169
3	L	W	$\frac{2}{3}W$	$\frac{2}{3}W$	0.375	0.431
4	L	L	$\frac{1}{3}W$	W	0.000	0.083
MSE probability	0.375	0.250	0.375	0.000		
Actual frequency	0.383	0.308	0.270	0.040		

Source: Mookerjee and Sopher (1997).

Note: L, W, pW denote loss, win, and a p chance of win.

13

Mookerjee and Sopher (1997)

Table 3.7. Row payoffs and results in games 2 and 4

Row choice	Column choice						MSE probability	Actual frequency
	1	2	3	4	5	6		
1	W	L	L	L	L	W	0.375	0.410
2	L	L	W	W	W	W	0.250	0.241
3	L	W	L	L	W	L	0.125	0.048
4	L	W	W	L	L	L	0.125	0.069
5	L	W	L	W	L	W	0.125	0.195
6	L	L	W	L	W	W	0.000	0.038
MSE probability	0.375	0.250	0.125	0.125	0.125	0.000		
actual frequency	0.368	0.269	0.099	0.166	0.080	0.019		

Source: Mookerjee and Sopher (1997).

14

Binmore, Swierzbinski, and Proulx (2001)

- Equilibrium play can only come about from learning, which implies that equilibration necessarily generates temporal dependence of the sort Brown and Rosenthal pointed out.
- Dependence is therefore evidence of equilibration.
- players play for 150 periods and can track moving averages of the payoffs to themselves and others
- Since they observe some convergence toward MSE over time, Camerer only reports frequencies from the last third of the experiment in the following tables.

15

Binmore et al. (2001)
Table 3.9. Game 1–5 row payoffs and results

	Column choice		MSE probability	Actual frequency
Row choice	1	1		
<i>Game 1</i>				
1	−2	3	0.167	0.251
2	−1	−2	0.833	0.749
MSE probability	0.833	0.167		
Actual frequency	0.915	0.085		

	Column choice			MSE probability	Actual frequency
Row choice	1	2	3		
<i>Game 2</i>					
1	−3	−2	−3	0.000	0.044
2	−1	−1	0	1.000	0.888
3	3	−3	−3	0.000	0.068
MSE probability	0.000	1.000	0.000		
Actual frequency	0.011	0.918	0.071		

16

Binmore et al. (2001)

Row choice	Column choice			MSE probability	Actual frequency
	1	2	3		
<i>Game 3</i>					
1	-2	3	-3	0.167	0.205
2	-1	-3	0	0.000	0.056
3	0	-1	1	0.833	0.739
MSE probability	0.667	0.333	0.000		
Actual frequency	0.647	0.279	0.075		

17

Binmore et al. (2001)

Table 3.9. (continued)

	Column choice			MSE probability	Actual frequency
Row choice	1	2	3		
<i>Game 4</i>					
1	0	2	-1	0.167	0.207
2	2	0	-1	0.167	0.133
3	-1	-1	0	0.667	0.660
MSE probability	0.167	0.167	0.667		
Actual frequency	0.081	0.171	0.748		

	Column choice				MSE probability	Actual frequency
Row choice	1	2	3	4		
<i>Game 5</i>						
1	1	-1	-1	-1	0.400	0.439
2	-1	-1	1	1	0.200	0.187
3	-1	1	-1	1	0.200	0.224
4	-1	1	1	-1	0.200	0.150
MSE probability	0.400	0.200	0.200	0.200		
actual frequency	0.448	0.266	0.112	0.174		

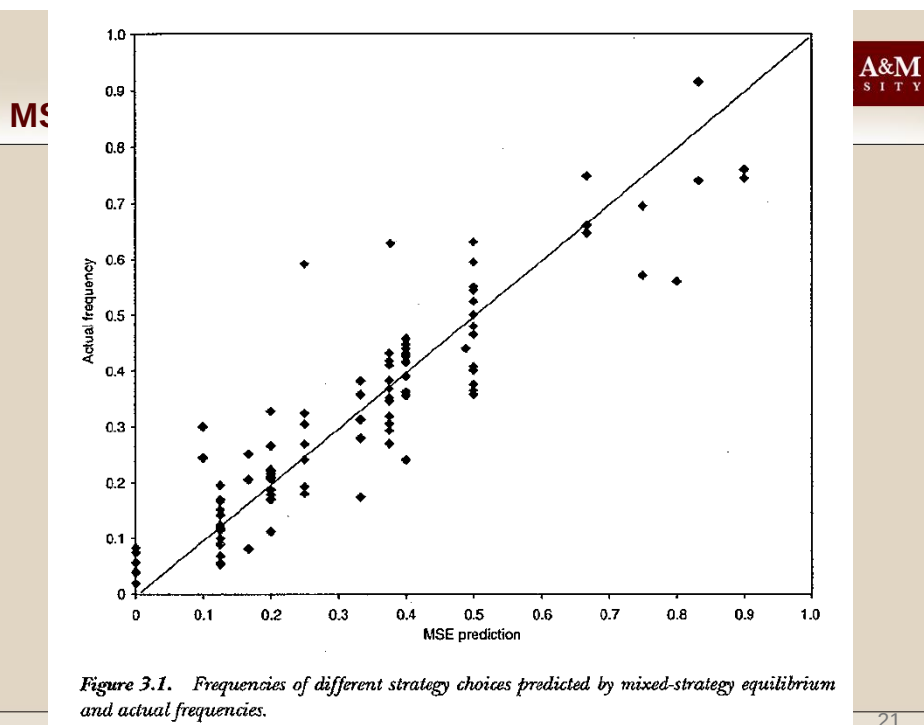
Source: Binmore, Swierzbinski, and Proulx (2001).

18

Camerer's Summary

- Modern studies have all used pairs of subjects playing against each other, for modest to large stakes and only win-loss payoffs so risk tastes can't possibly matter
- Actual frequencies are not far from the MSE predictions.
- Deviations were the smallest in the experiments by Binmore et al. in which players play for 150 periods and can track moving averages of the payoffs of themselves and others.
- However, deviations are usually statistically significant
- Most subjects exhibit temporal dependence

20



21

Logis

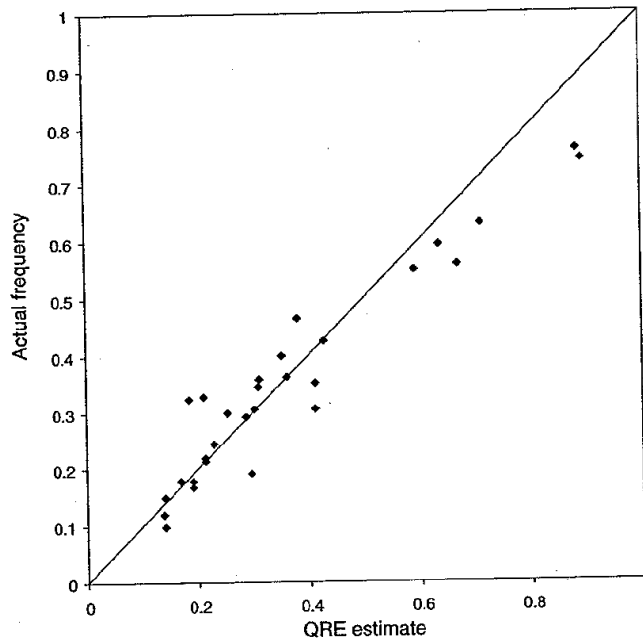


Figure 3.3. Frequencies of different strategy choices predicted by quantal response equilibrium and actual frequencies.

22



Field Studies

- Walker and Wooders (2001) Tennis serves
 - Ten matches between famous tennis pros
 - 1974-1997
 - In a mixed equilibrium, the expected payoff from serving left (L) or right (R) should be equal.
 - If the payoffs from L and R are different, then the players are not mixing properly.
 - Each match has two players serving from two different sides of the court, so there are 40 comparisons.
 - The L and R winning frequencies are statistically different (at 10 percent level) in only two of the forty comparisons

23

Palacios-Heurta (2001)

- Penalty kicks in European soccer

Table 3.15. *Scoring rates for left and right moves in soccer*

	Goalie move direction		
Kick direction	L	R	Overall
<i>Left-footed kickers</i>			
L	0.62	0.95	0.76
R	0.94	0.61	0.81
Overall	0.76	0.80	0.78
<i>Right-footed kickers</i>			
L	0.50	0.94	0.76
R	0.98	0.73	0.83
Overall	0.77	0.82	0.80